

ELEC4631 Lecture 5: Lyapunov Stability Analysis Part 1

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Outline

- Internal stability for dynamical systems.
- Stability analysis with Lyapunov functions.

Additional reference

- A scanned copy of Chapter 3 (Fundamentals of Lyapunov Theory) of the textbook of Slotine and Li is available on the course Moodle page.
- This provides additional reading material and exercise to complement lecture notes 5 and 6 and the associated tutorials

Internal stability

- In dynamical systems, state typically represents internal variables of system. They must be stable.
- Stable means no element of state has magnitude that grows over time.
- Such stability often referred to as *internal stability*. Stability absolutely critical -- any control system design must ensure stability.

Internal stability

- There can be different notions of internal stability. Here we look at
 - Stability in the sense of Lyapunov (i.s.L).
 - Asymptotic stability.
 - Global asymptotic stability.

Equilibrium point

- For ODE

$$\dot{x}(t) = f(x(t))$$

a state x^* in S is said to be an *equilibrium point* if $f(x^*) = 0$. It is a point where vector field vanishes.

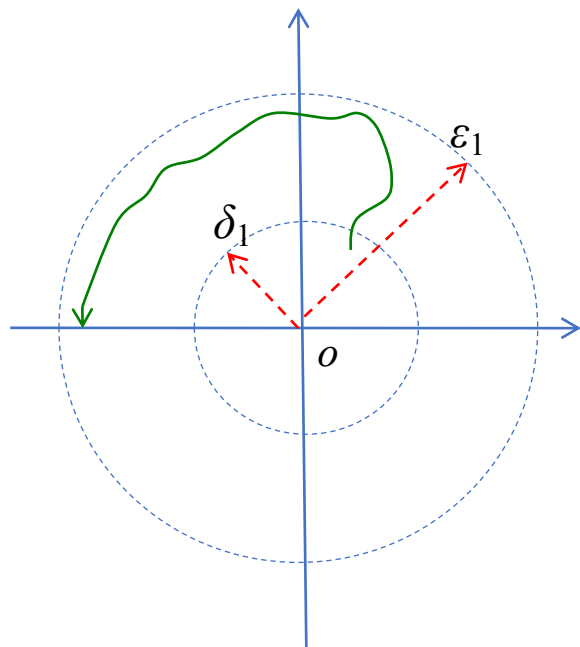
- For such point we have that if $x(t_0) = x^*$ at $t = t_0$ then $x(t) = x^*$ for all $t > t_0$.
- That is, if a system's state at some time is x^* it remains there for all times after.
- System may have more than one equilibrium point.

Stability in the sense of Lyapunov

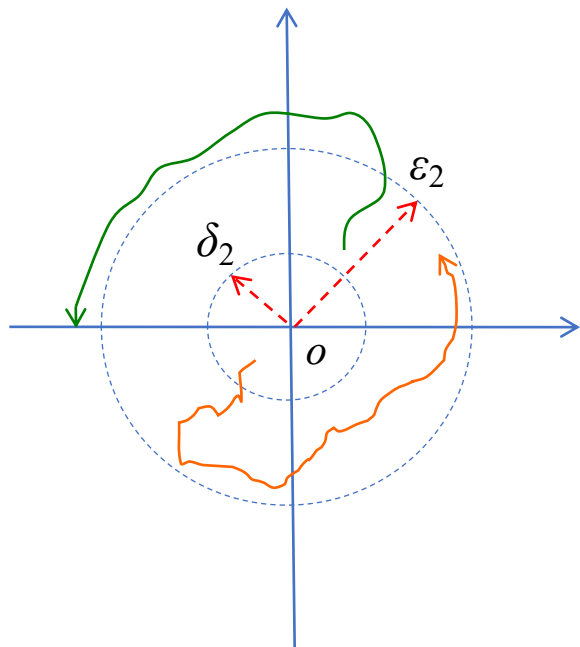
- A continuous-time dynamical system with $f(x^*) = 0$ is *stable in the sense of Lyapunov* (i.s.L) around x^* if for any $\varepsilon > 0$ there exists a $\delta > 0$ (depending on ε) such that $\|x(0) - x^*\| < \delta$ implies $\|x(t) - x^*\| < \varepsilon$ for all $t \geq 0$.
- Here $\|x(t)\|$ is Euclidean norm of $x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$:

$$\|x(t)\| = \sqrt{\sum_{k=1}^n |x_k(t)|^2}$$

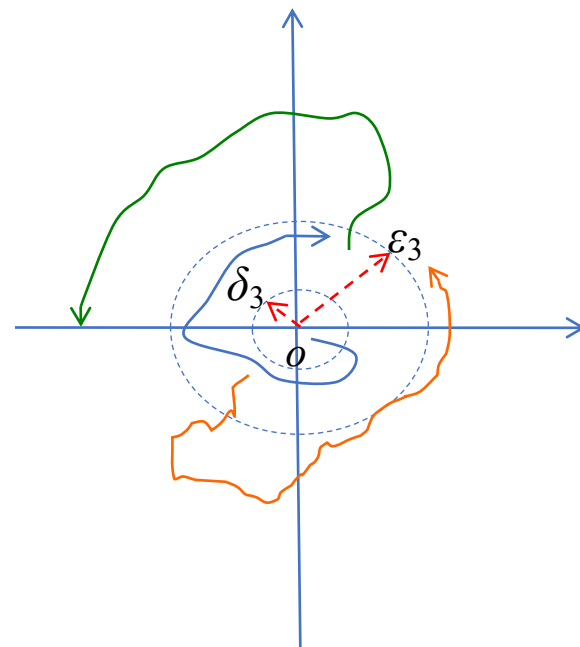
Stability in the sense of Lyapunov



$$x^* = 0$$



$$\varepsilon_3 < \varepsilon_2 < \varepsilon_1$$



$$\delta_3 < \delta_2 < \delta_1$$

Stability in the sense of Lyapunov

- Consider the RLC circuit with no resistance ($R = 0$), $L = 1$ H, and $C = 1$ F, so is LC circuit.
- One equilibrium point: $x^* = (0,0)$.
- Solution of ODE is

$$v_C(t) = \sqrt{i_0^2 + v_0^2} \sin(t + \phi)$$

$$i(t) = \sqrt{i_0^2 + v_0^2} \cos(t + \phi)$$

$$\phi = \tan^{-1} \left(\frac{v_0}{i_0} \right)$$

Stability in the sense of Lyapunov

- State of LC circuit is $x(t) = (v_C(t), i(t))^T$.
- If we want

$$\|x(t) - x^*\| = \sqrt{v_C(t)^2 + i(t)^2} < \varepsilon \text{ for all } t \geq 0$$

it is enough to start at initial state (v_0, i_0) satisfying

$$\|(v_0, i_0)^T - x^*\| = \sqrt{v_0^2 + i_0^2} < \delta$$

with $\delta = \varepsilon$.

- LC circuit stable i.s.L around equilibrium point $x^* = (0,0)$.

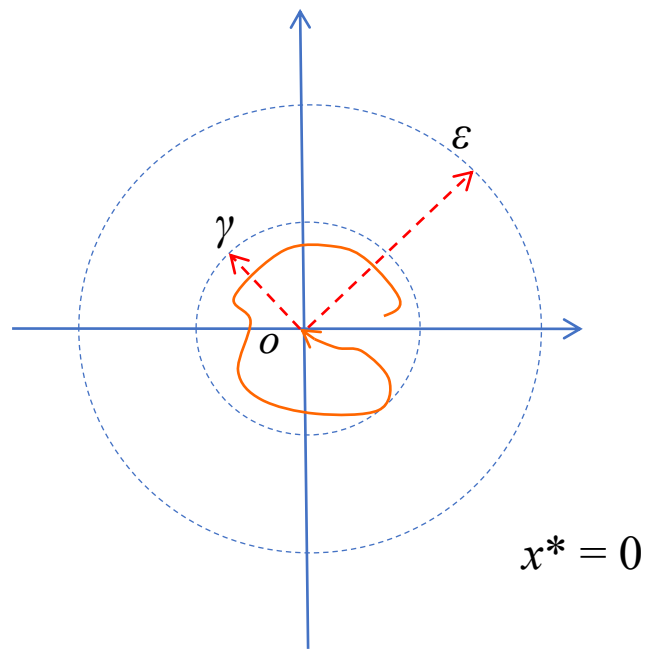
Asymptotic stability

- A continuous-time dynamical system with $f(x^*) = 0$ is *asymptotically stable* around x^* if (i) it is stable i.s.L *and* (ii) there exists $\gamma > 0$ such that if $\|x(0) - x^*\| < \gamma$ then $x(t) \rightarrow x^*$ as $t \rightarrow \infty$.

Asymptotic stability

- What this says: if system is asymptotically stable around x^* then (i) it is stable i.s.L around x^* and (ii) you can guarantee that $x(t)-x^*$ goes to origin as $t \rightarrow \infty$ if you start system at initial state $x(0)$ lying in a multidimensional ball of radius γ around x^* .

Asymptotic stability



Asymptotic stability

- Consider the RLC circuit with $R = 1 \text{ } \Omega$, $L = 0.5 \text{ H}$, and $C = 1 \text{ F}$.
- Only one equilibrium point: $x^* = (0,0)$.
- Solution of ODE is

$$v_C(t) = \sqrt{2v_0^2 + 2v_0i_0 + i_0^2} e^{-t} \cos(t + \phi)$$

$$i(t) = -\sqrt{2v_0^2 + 2v_0i_0 + i_0^2} e^{-t} (\cos(t + \phi) + \sin(t + \phi))$$

$$\phi = -\arctan\left(\frac{v_0 + i_0}{v_0}\right)$$

Asymptotic stability

- So, if we want

$$\|x(t) - x^*\| < \varepsilon \text{ for all } t \geq 0$$

it is enough to start at initial state (v_0, i_0) satisfying

with $\delta = \varepsilon/(2\sqrt{3})$.
$$\|(v_0, i_0)^T - x^*\| < \delta$$

- RLC circuit is stable in the sense of Lyapunov.

Asymptotic stability

- Moreover, for any initial state (v_0, i_0) ,

$$\begin{aligned}\lim_{t \rightarrow \infty} v_C(t) &= \lim_{t \rightarrow \infty} \sqrt{2v_0^2 + 2v_0i_0 + i_0^2} e^{-t} \cos(t + \phi) \\ &= 0\end{aligned}$$

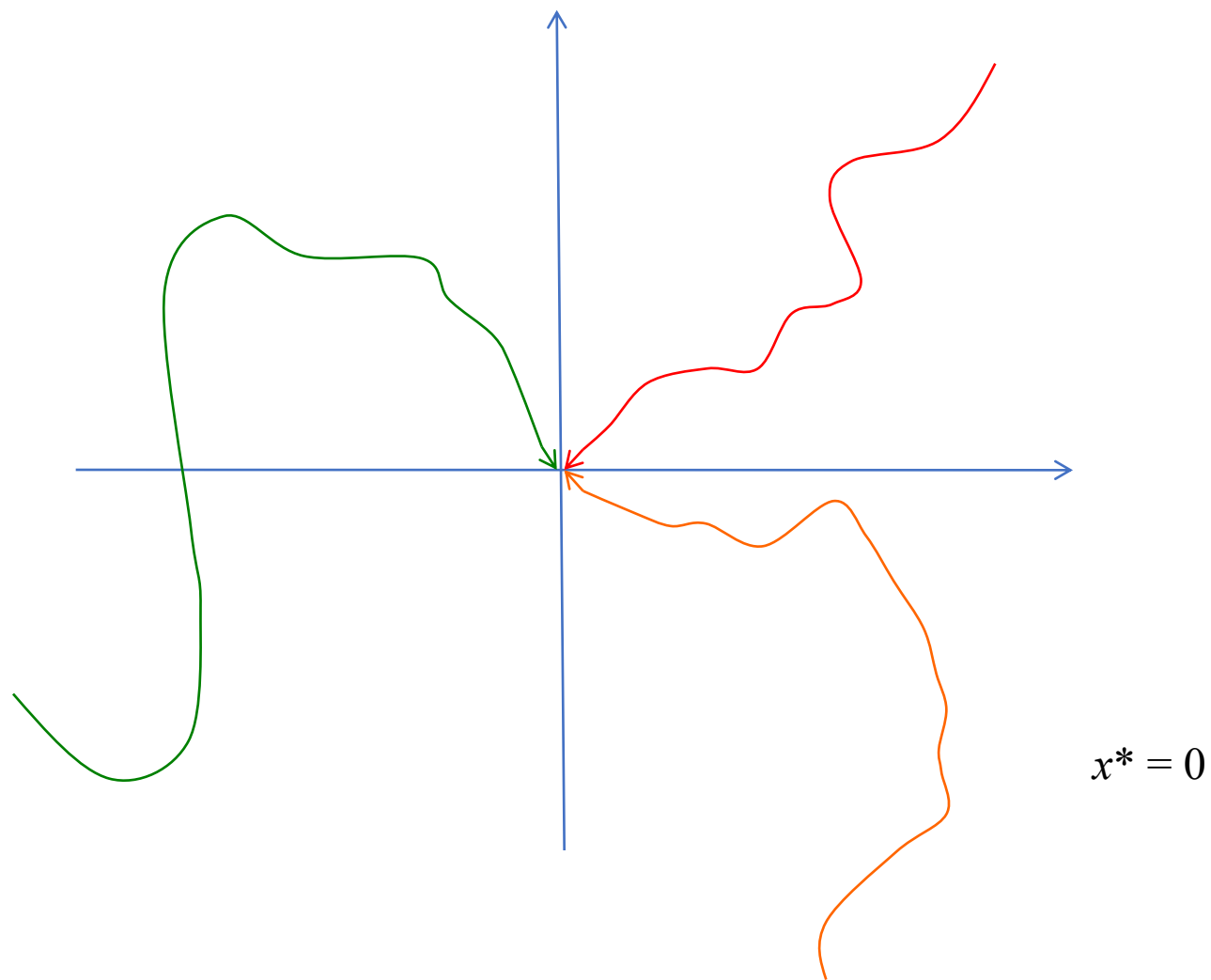
$$\begin{aligned}\lim_{t \rightarrow \infty} i(t) &= \lim_{t \rightarrow \infty} -\sqrt{2v_0^2 + 2v_0i_0 + i_0^2} e^{-t} (\cos(t + \phi) + \sin(t + \phi)) \\ &= 0\end{aligned}$$

- Thus γ can be chosen to be any positive constant.
- This RLC circuit asymptotically stable around equilibrium point $x^* = (0,0)$.

Global asymptotic stability

- A continuous-time dynamical system with $f(x^*) = 0$ is *globally asymptotically stable* around x^* if $x(t) \rightarrow x^*$ as $t \rightarrow \infty$ starting from *any* initial state $x(0)$.
- Of course, if x^* is globally asymptotically stable then it must be the *only* equilibrium point of the system.

Global asymptotic stability



Global asymptotic stability

- For RLC circuit with $R = 1 \text{ } \Omega$, $L = 0.5 \text{ H}$, and $C = 1 \text{ F}$, $v_C(t) \rightarrow 0$ and $i(t) \rightarrow 0$ as $t \rightarrow \infty$ for any initial state (v_0, i_0)
- This RLC circuit besides asymptotically stable is also globally asymptotically stable around equilibrium point $x^* = (0,0)$.

Determining stability is difficult

- Determining stability of continuous-time dynamical system is in general difficult.
- Why difficult? Because in general not possible to explicitly solve given ODE

$$\dot{x}(t) = f(x(t)), \quad x(0) = x_0$$

to express $x(t)$ as function of initial state x_0 and time t , i.e., as $x(t) = R(t, x_0)$.

- One exception is the case where f is linear (covered in later weeks).

Stability analysis with Lyapunov functions

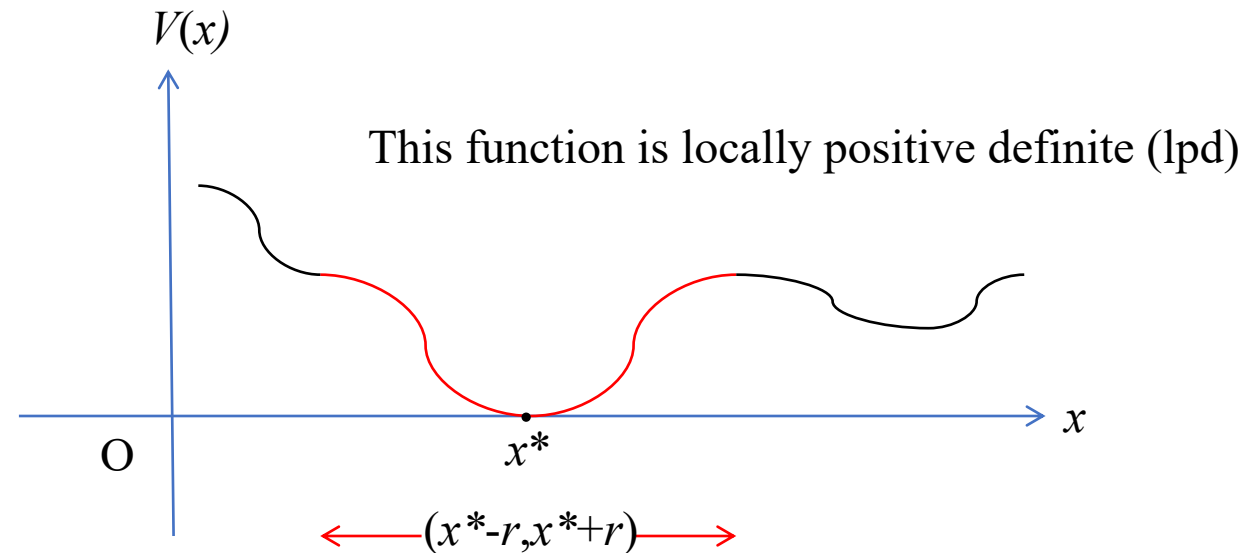
- Russian mathematician Aleksandr Lyapunov devised a very original method to analyse stability without solving ODE.
- Done with help of so-called *Lyapunov functions*.
- The method is called *second* (or *direct*) *method of Lyapunov* (first method of Lyapunov will be covered in Week 7).

Locally positive definite functions

- Let state-space be Euclidean space $\mathbb{R}^n$. A function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be *locally positive definite* (lpd) around x^* if
 1. V is *continuous* and $V(x^*) = 0$.
 2. $V(x) > 0$ if $0 < \|x - x^*\| < r$ for some $r > 0$.
- V is said to be *locally positive semidefinite* (lpsd) around x^* if $V(x) > 0$ in (2) is replaced by $V(x) \geq 0$ and for any $r > 0$ there is $x \neq x^*$ with $\|x - x^*\| < r$ such that $V(x) = 0$.

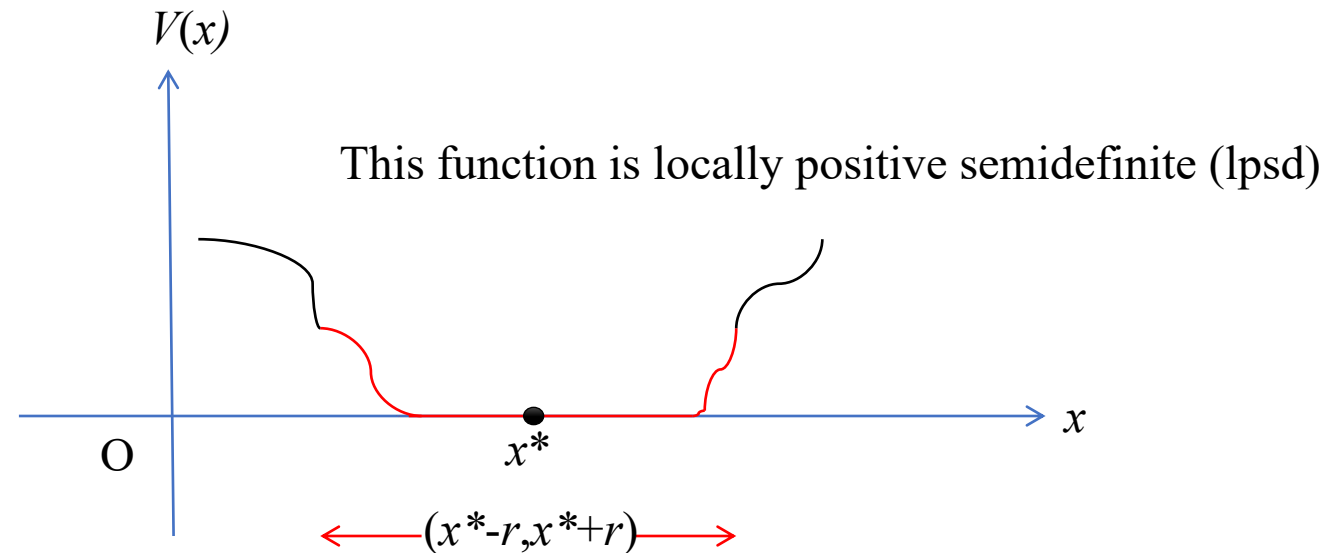
Locally positive definite functions

- Consider a system with 1D state-space and equilibrium point x^*



Locally positive semidefinite functions

- Consider a system with 1D state-space and equilibrium point x^*



Locally negative definite and negative semidefinite functions

- A function V is said to be locally negative definite (lnd) (resp., locally negative semi-definite (lnsd)) around x^* if $-V$ is lpd (resp., lpsd) around x^* .

Globally definite and semidefinite functions

- If V satisfies $V(x^*) = 0$ and $V(x) > 0$ for all $x \neq x^*$ then the function is said to be *positive definite* (pd) *around* x^*
- V is negative definite (nd) around x^* if $-V$ is pd around x^*
- If $x^* = 0$ then we simply say that V is pd or nd (no need to say around 0 in this case)

Globally definite and semidefinite functions

- A function V is positive semidefinite (psd) around x^* if $V(x^*) = 0$ and $V(x) \geq 0$ for all x
- A function V is negative semidefinite (nsd) around x^* if $-V$ is psd around x^*
- If $x^* = 0$ then we simply say that V is psd or nsd (no need to say around 0 in this case)

Some subtleties with general semidefinite functions

- A function that is pd (nd) around x^* is automatically also lpd (lnd) around x^* .
- However, a function that is psd around x^* may either be lpd or lpsd around x^* , depending on where it is zero at other points other than x^* .
- Similarly, a function that is nsd around x^* may either be lnd or lnsd around x^* , depending on where it is zero at other points other than x^* .
- So, a function can be simultaneously psd and lpd, or psd and lpsd. Similarly, a function can be nsd and lnd, or nsd and lnsd
- Some examples will follow.

Example 1: A function psd around x^* that is lpd around x^* but not lpsd around x^*

- Consider the function $f(x) = x^2 \sin^2(x)$ on $\mathbb{R}$. This function is psd around $x^* = n\pi$ for any integer n since it is clearly zero at x^* and is non-negative (≥ 0) everywhere else.
- However, this function is lpd around $x^* = n\pi$ since in the interval $(x^* - \pi, x^* + \pi)$ it can be checked that $f(x) > 0$ for all points in this interval except at $x = x^*$
- The function f is *not* lpsd around x^*

Example 2: A function psd around x^* which is lpsd around x^*

- Consider the function $f(x,y) = (x-y)^2$ on $\mathbb{R}^2$. This function is psd around $(x^*,y^*) = (0,0)$ since it is clearly zero at (x^*,y^*) and is non-negative everywhere else.
- However, this function is lpsd around $(x^*,y^*) = (0,0)$ since for *any* $r > 0$ there always exists a point $(x,y) \neq (0,0)$ with $\|(x,y)\| < r$ such that $f(x,y) = 0$. That is, take any point (a,a) with $0 < |a| < r/\sqrt{2}$
- This function is *not* lpd around $(0,0)$

Some subtleties with general semidefinite functions

- It is crucial to internalise the different notions of definiteness, the subtleties and be comfortable with them
- Make sure you fully understand the difference between local notions (lpd, lnd, lpsd, lnsd) vs global ones (pd, nd, psd, nsd) around x^*

Lyapunov functions

- We also assume that V has continuous partial derivatives.
- That is, $\frac{\partial V}{\partial x_i}$ is continuous for $i = 1, 2, \dots, n$.

Lyapunov functions

- Recall that

$$\begin{aligned}\dot{V}(x(t)) &= \frac{dV(x(t))}{dt} \\ &= \sum_{i=1}^n \frac{\partial V}{\partial x_i}(x(t)) \dot{x}_i(t), \\ &= \sum_{i=1}^n \frac{\partial V}{\partial x_i}(x(t)) f_i(x(t)), \\ &= \nabla V(x(t))^T f(x(t)),\end{aligned}\quad f(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_n(x) \end{bmatrix}$$

where

$$\nabla V(x(t))^T = \left[\frac{\partial V}{\partial x_1}(x(t)) \quad \frac{\partial V}{\partial x_2}(x(t)) \quad \dots \quad \frac{\partial V}{\partial x_n}(x(t)) \right].$$

Lyapunov functions

- Lyapunov function: Let $f(x^*) = 0$. If V is lpd around x^* and

$$\dot{V}(x) = \nabla V(x)^T f(x)$$

is lnsd *or* lnd around x^* then V is said to be a Lyapunov function around x^* .

- If V is pd around x^* and $\dot{V}$ is nsd or nd around x^* then V is said to be a global Lyapunov function.

Radially unbounded functions

- A scalar function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be *radially unbounded* if g is continuous and $|g(x)| \rightarrow \infty$ as $\|x\| \rightarrow \infty$.

Second method of Lyapunov

- **Theorem:** Let x^* be an equilibrium point, $f(x^*) = 0$.

1. If there exists a Lyapunov function V around x^* with

$$\dot{V}(x) = \nabla V(x)^T f(x)$$

lnd around x^* then x^* is stable i.s.L.

2. If there exists a Lyapunov function V around x^* such that $\dot{V}$ is lnd around x^* then x^* is an asymptotically stable equilibrium point.
3. If V is pd around x^* , $\dot{V}$ is nd around x^* and V is radially unbounded, then x^* is globally asymptotically stable.

Second method of Lyapunov

- What theorem says: if one can find a Lyapunov function that has properties stated in the theorem, it is possible to determine whether an equilibrium point x^* is stable i.s.L, asymptotically stable, or globally asymptotically stable around x^* , without solving ODE!

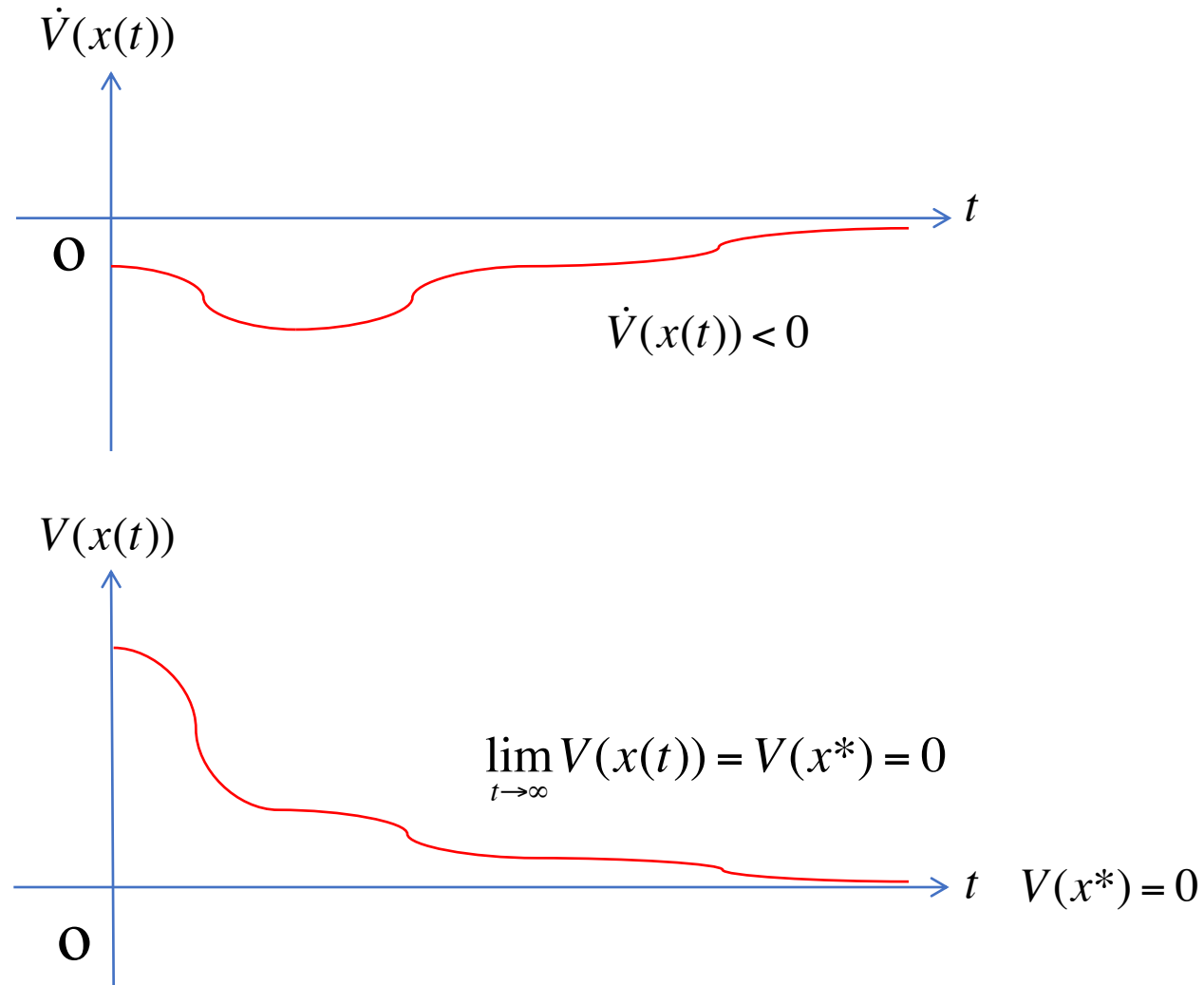
Intuition behind second method of Lyapunov

- We will not prove this theorem, but there is a clear intuition behind it coming from physics ideas.
- Gist of it: Lyapunov function V can be thought of as playing the role of an abstract energy-like function for system (but not necessarily energy in the strict sense of physics).

Intuition behind second method of Lyapunov

- When $\dot{V}(x(t)) < 0$, “energy” is continuously decreasing along any trajectory until it reaches the point x^* where energy is 0 (since $V(x^*) = 0$). At that point system comes to rest because all energy has been expended.

Derivative of $V(x(t))$ ind, $x(t)$ converges to x^*
as $t \rightarrow \infty$



Example 1

- Consider a system given by the nonlinear ODE

$$\dot{x}_1 = x_1(x_1^2 + x_2^2 - 2) - 4x_1x_2^2$$

$$\dot{x}_2 = 4x_1^2x_2 + x_2(x_1^2 + x_2^2 - 2)$$

- The vector field here is

$$f(x_1, x_2) = \begin{bmatrix} x_1(x_1^2 + x_2^2 - 2) - 4x_1x_2^2 \\ 4x_1^2x_2 + x_2(x_1^2 + x_2^2 - 2) \end{bmatrix}$$

Example 1

- The origin is an equilibrium of the system since $f(0,0) = 0$. Are there other equilibria? Find out as an exercise!
- Is the origin an asymptotically stable equilibrium?
- Consider the candidate Lyapunov function

$$V(x_1, x_2) = x_1^2 + x_2^2$$

Example 1

- Candidate function has the property $V(0,0) = 0$ and $V(x_1, x_2) > 0$ for all $(x_1, x_2) \neq (0,0)$. It is pd and hence also lpd around the origin
- Compute $\dot{V}(x)$:

$$\begin{aligned}\dot{V}(x) &= \nabla V(x)^T f(x) \\ &= 2(x_1^2 + x_2^2)(x_1^2 + x_2^2 - 2)\end{aligned}$$

Example 1

- An alternative (sometimes quicker) way to compute $\dot{V}(x)$:

$$\begin{aligned}\dot{V}(x) &= 2(x_1\dot{x}_1 + x_2\dot{x}_2) \\ &= 2(x_1^2 + x_2^2)(x_1^2 + x_2^2 - 2)\end{aligned}$$

verify the calculations as an exercise

- We have that $\dot{V}(0,0) = 0$ and $\dot{V}(x) < 0$ for all x satisfying

$$0 < \|x\| = \sqrt{x_1^2 + x_2^2} < \sqrt{2}$$

Example 1

- Therefore, $\dot{V}$ is lnd around (0,0) (but it is not nd!)
- Since
 - V is lpd around (0,0) (since it is pd) and
 - $\dot{V}$ is lnd around (0,0)

We conclude from the theorem of Lyapunov's second method that (0,0) is an *asymptotically stable* (A.S) equilibrium

Example 2

- Consider a system described by the ODE

$$\dot{x} = \sin^2(x) - x$$

- The vector field here is

$$f(x) = \sin^2(x) - x$$

- The origin is an equilibrium of the system since $f(0) = 0$.

Example 2

- Consider the candidate Lyapunov function

$$V(x) = x^2$$

- This candidate function is pd and is radially unbounded, $V(x) \rightarrow \infty$ as $|x| \rightarrow \infty$
- Computing $\dot{V}$ we get

$$\dot{V}(x) = -2x(x - \sin^2(x))$$

- This $\dot{V}$ is nd (show this as an exercise)

Example 2

- Since
 - V is pd and
 - $\dot{V}$ is nd and
 - V is radially unbounded

We conclude by the theorem of Lyapunov's second method that 0 is a *globally asymptotically stable* (G.A.S) equilibrium

Example 2

- Note that we did not check if 0 is the only equilibrium of the system
- However, because we have established that the origin is G.A.S by the second method of Lyapunov, the origin *must be* the only equilibrium